Resumes: Temperance — Answers

Temperance

Predictions

Use `predictions(fit_resumes, type = "prob")` to estimate the callback probability for individual resumes in the sample.

```r
predictions(fit_resumes, type = "prob") |>
  as_tibble() |>
  filter(group == "yes") |>
  select(ethnicity, city, sex, special, estimate, conf.low, conf.high) |>
  head(6)
```

```
# A tibble: 6 x 7
  ethnicity city    sex    special estimate conf.low conf.high
  <chr>     <chr>   <chr>  <chr>      <dbl>    <dbl>     <dbl>
1 cauc      chicago female no        0.0594   0.0476    0.0712
2 cauc      chicago female no        0.0594   0.0476    0.0712
3 afam      chicago female no        0.0388   0.0302    0.0475
4 afam      chicago female yes       0.0838   0.0668    0.101
5 cauc      chicago female no        0.0594   0.0476    0.0712
6 cauc      chicago male   yes       0.112    0.0809    0.143
```

`predictions()` returns one row per resume *per outcome class*; we filter to `group == "yes"` because the callback probability is what we care about. Each resume gets its own predicted probability, built from its own covariates — this is the stepping stone, not the answer.

Use `avg_predictions(fit_resumes, type = "prob", by = "ethnicity")` to estimate the average callback probability for resumes with black-sounding and white-sounding names.

```r
avg_predictions(fit_resumes, type = "prob", by = "ethnicity")
```

```
 Group ethnicity Estimate Std. Error     z Pr(>|z|)     S  2.5 % 97.5 %
   no       afam   0.9355    0.00494 189.3   <0.001   Inf 0.9258 0.9452
   no       cauc   0.9035    0.00593 152.5   <0.001   Inf 0.8919 0.9151
   yes      afam   0.0645    0.00494  13.0   <0.001 126.8 0.0548 0.0742
   yes      cauc   0.0965    0.00593  16.3   <0.001 195.7 0.0849 0.1081

Type: prob
```

Read the `yes` rows: these are the headline numbers — the average callback probability for each kind of name, with confidence intervals. (The `no` rows are their mirror images.)

What is the expected callback probability for an applicant with a black-sounding name (with 95% CI)?

> About 6.5%, with a 95% CI of roughly [5.5%, 7.4%].

What is the expected callback probability for an applicant with a white-sounding name (with 95% CI)?

> About 9.7%, with a 95% CI of roughly [8.5%, 10.8%].

Comparisons

Use `avg_comparisons(fit_resumes, variables = "ethnicity", type = "prob")` to estimate the difference in callback probability between white-sounding and black-sounding names.

```r
avg_comparisons(fit_resumes, variables = "ethnicity", type = "prob")
```

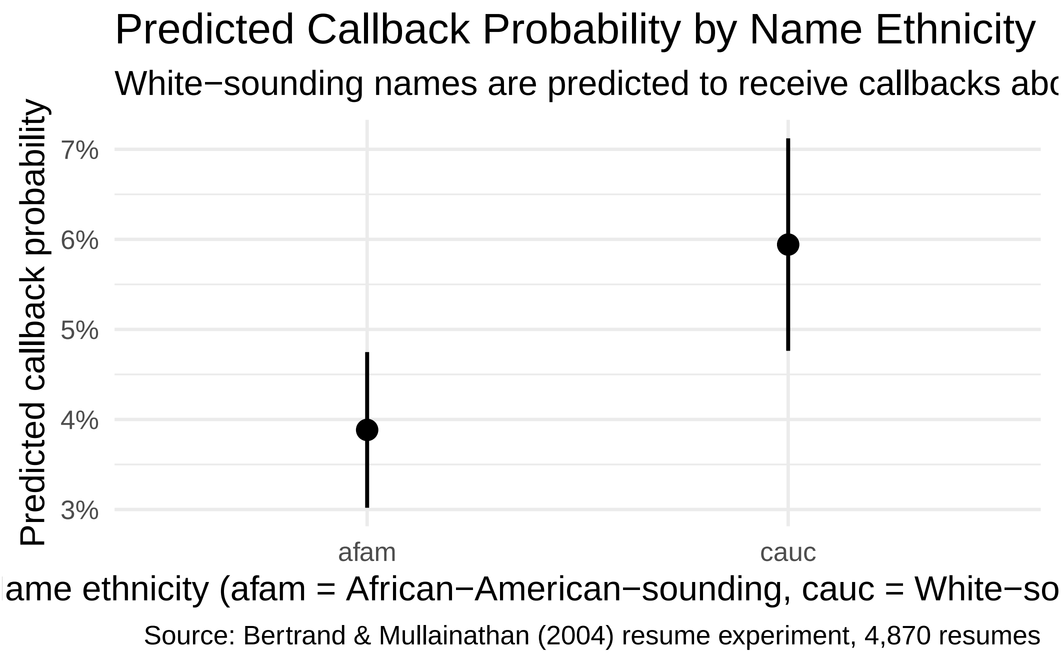
```
 Group Estimate Std. Error     z Pr(>|z|)    S   2.5 %  97.5 %
   no   -0.0321    0.00772 -4.16   <0.001 15.0 -0.0473 -0.0170
   yes   0.0321    0.00772  4.16   <0.001 15.0  0.0170  0.0473

Term: ethnicity
Type: prob
Comparison: cauc - afam
```

The `yes` row is the single number the questions ask for: a white-sounding name is associated with a callback probability about 3.2 percentage points higher (95% CI roughly [1.7, 4.7]) than a black-sounding name. On a base rate of 6.5%, 3.2 points is enormous — about 50% more callbacks. In practical terms, an applicant with a black-sounding name must send roughly 15 resumes to expect one callback; an identical resume with a white-sounding name needs about 10.

Visualizations

Create a visualization of the average predictions by ethnicity using `plot_predictions()`.

```
plot_predictions(fit_resumes, condition = "ethnicity", type = "prob", draw = FALSE) |>
  filter(group == "yes") |>
  ggplot(aes(x = ethnicity, y = estimate)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high), linewidth = 0.7) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Predicted Callback Probability by Name Ethnicity",
    subtitle = "White-sounding names are predicted to receive callbacks about 3 percentage po
    x = "Name ethnicity (afam = African-American-sounding, cauc = White-sounding)",
    y = "Predicted callback probability",
    caption = "Source: Bertrand & Mullainathan (2004) resume experiment, 4,870 resumes"
  ) +
  theme_minimal(base_size = 13)
```



The plot makes the size of the gap — and the precision with which we estimate it — visible at a glance: the two intervals are far from overlapping.

What measure of uncertainty is shown in this plot?

> The vertical bars are 95% confidence intervals around each group's predicted average callback probability.

Interpretation

How does the historian interpret the difference in callback probabilities? How does the civil rights organization interpret the same number differently?

> **Historian:** the 3.2-point gap is a *comparison across rows* — applicants with black-sounding names got callbacks less often than applicants with white-sounding names. It describes the process that sorted applicants into jobs; it makes no claim about what would have happened to any particular applicant under a different name.
>
> **Civil rights organization:** the same 3.2 points is an estimate of a *within-row* quantity — the average difference between each application's two potential outcomes. Because names were randomly assigned, the across-group comparison identifies the average causal effect: putting a black-sounding name on an otherwise identical resume *causes* roughly a one-third drop in the chance of a callback.

Humility

Besides the average callback probabilities, identify two additional quantities of interest the person in your scenario might care about.

> **Historian:** the callback gap by *sex* (her question names it directly, and our model's `sex` coefficient addresses it), and how the racial gap differed between cities — was Baltimore likely closer to the Boston pattern or the Chicago pattern?
>
> **Civil rights organization:** whether the causal effect varies with resume quality — if a stronger resume cannot close the gap (as the EDA hinted), that redirects policy from "help applicants build resumes" toward the employers doing the screening. Also the effect expressed in applications-per-callback (about 15 versus 10), which is the number a job-seeker actually experiences.

Why might these estimates be a poor answer to your scenario's question? Identify at least two possible sources of bias.

1) The data are 25 years old (or, for the historian, from different cities than Baltimore). If hiring discrimination has changed in level or form since 2001 — or differed between Baltimore and Boston/Chicago in 2000 — the stability and representativeness gaps discussed in Justice bias our answer in ways the confidence intervals do not capture.

2) The resumes were fictitious, sent only to newspaper help-wanted ads for entry-level jobs. Real applicant pools, real resumes, and modern online application channels may respond differently, so the estimated gap may not transfer to either protagonist's actual population of applications.

3) The world is always more uncertain than our models would have us believe: even with every assumption exactly right, the reported intervals are only the *Preceptor's Posterior* — the best posterior achievable under our assumptions — not the truth.